

ICT702 Data Visualisation

Selecting a Chart Type

Module 2





I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



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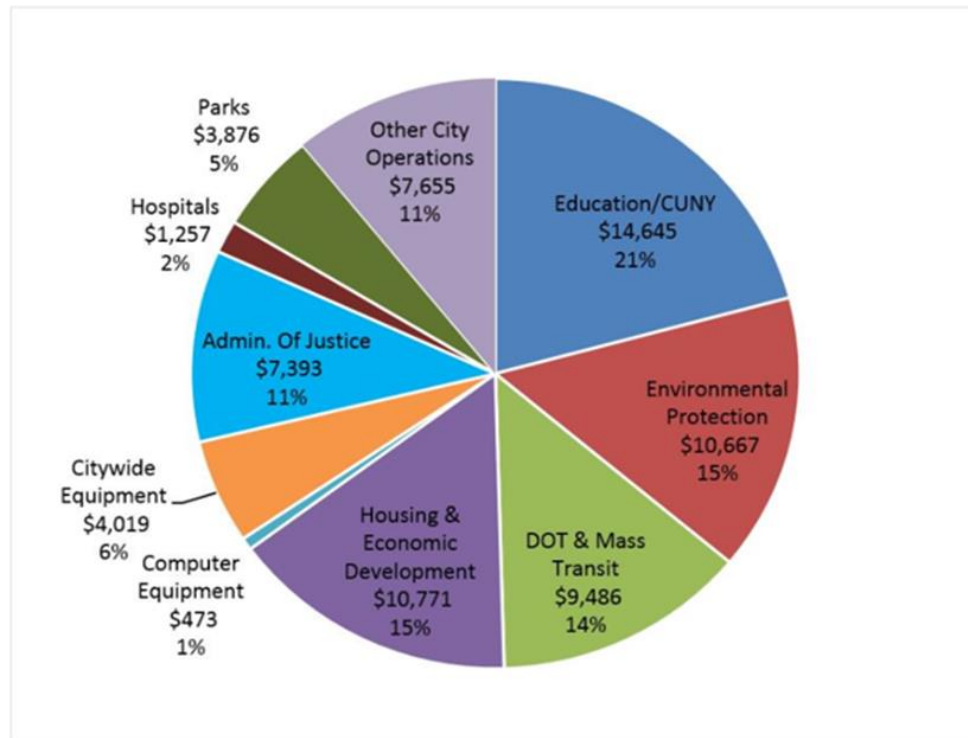
Learning Objectives for Module 2

- Create charts and graphs using Excel
- Edit charts and graphs created using Excel
- Select appropriate chart type for a given goal and data type
- Glean insights from charts and graphs
- Describe chart types to be avoided and why

Part 1

Data Visualisation Makeover

A pie chart showing the NYC budget allocation



Source: NYC Office of Management and Budget, FY 2020 Adopted Capital Commitment Plan, October 2019 Pie Chart figures may not tie due to rounding.

A bar chart showing the NYC budget allocation



Selecting an Appropriate Chart

Common goals for charts are to show:

- **Composition** – components
- **Ranking** – order
- **Relationship** – patterns
- **Distribution** – dispersion

Creating a Chart in Excel

	A	B
1	Month	Attendance
2	Jan	5422
3	Feb	4878
4	Mar	6586
5	Apr	6943
6	May	7876
7	Jun	17843
8	July	21967
9	Aug	14542
10	Sept	8751
11	Oct	6454
12	Nov	5677
13	Dec	11422

To create a column chart in Excel.

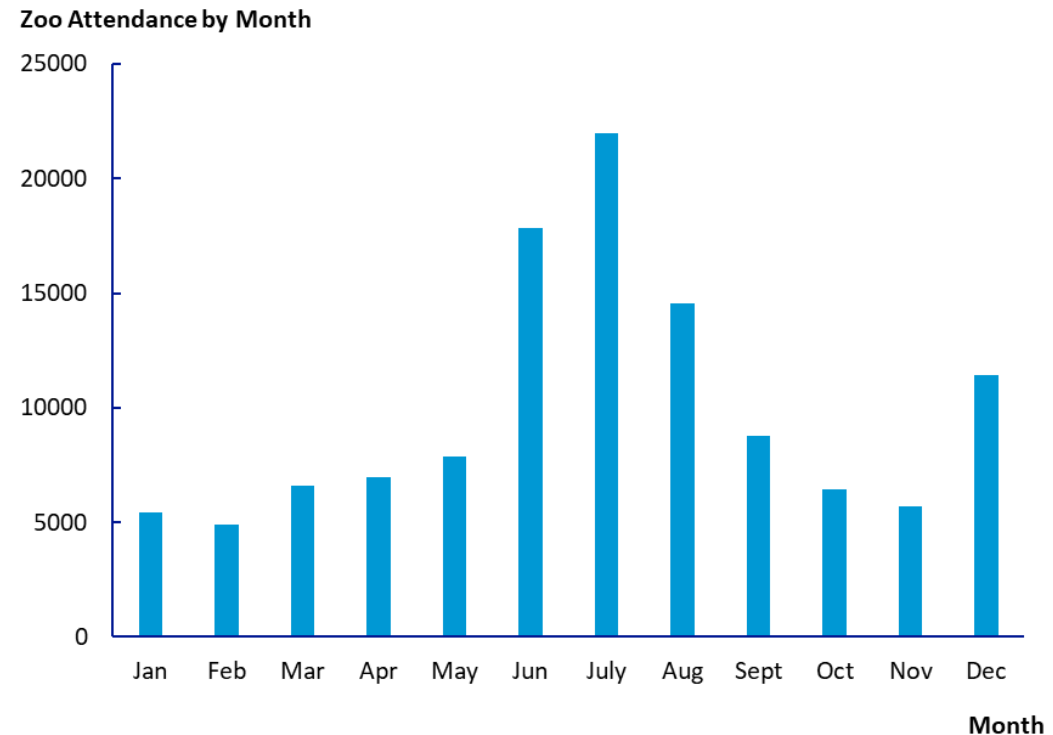
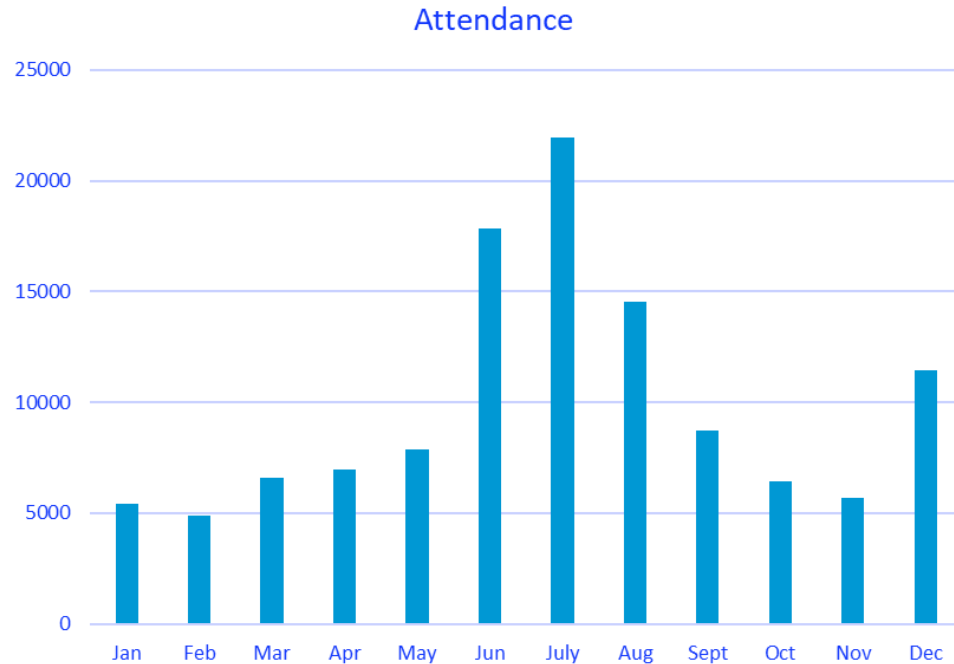
1. Select cells A1:B13
2. Click **Insert** tab on Ribbon
3. Click **Insert Column or Bar Chart** button in **Charts group**
4. Click **Clustered Column** button

In Excel:

- horizontal axis generally represents leftmost column of data.
- assignment of data to axes can be switched

Editing a Chart in Excel

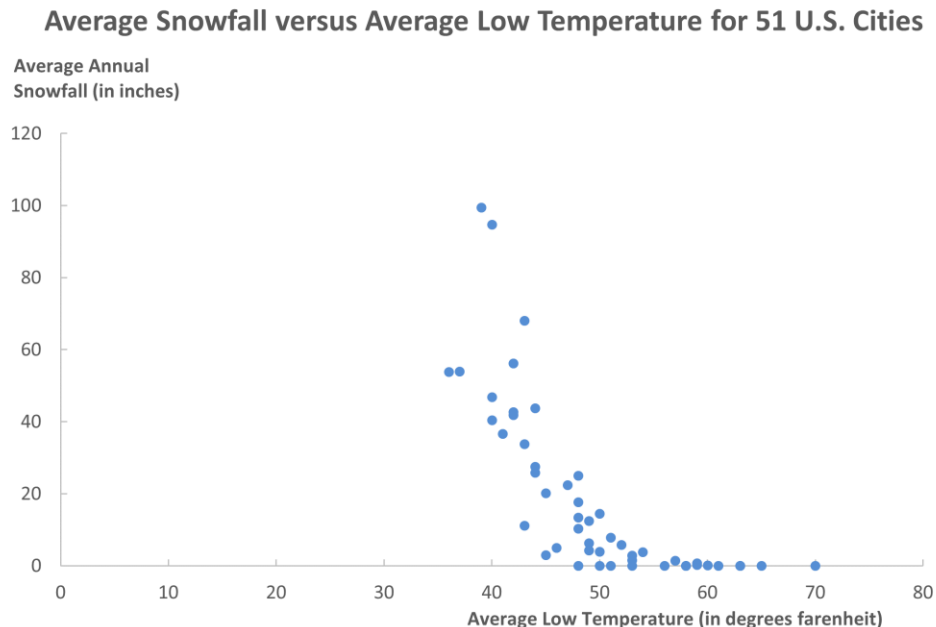
The chart shown on the right is clearer and easier to read than the chart shown on the left.



	A	B	C	D
1	City	State	Average Low temp	Average Snowfall
2	Atlanta	Georgia	53	2.9
3	Austin	Texas	59	0.6
4	Baltimore	Maryland	45	20.2
5	Birmingham	Alabama	53	1.6
6	Boston	Massachusetts	44	43.8
7	Buffalo	New York	40	94.7
8	Charlotte	North Carolina	49	4.3
9	Chicago	Illinois	41	36.7
10	Cincinnati	Ohio	43	11.2
11	Cleveland	Ohio	43	68.1
12	Columbus	Ohio	44	27.5
13	Dallas	Texas	57	1.5
14	Denver	Colorado	36	53.8
15	Detroit	Michigan	42	42.7
16	Hartford	Connecticut	40	40.5

Building and Interpreting a Scatter Chart

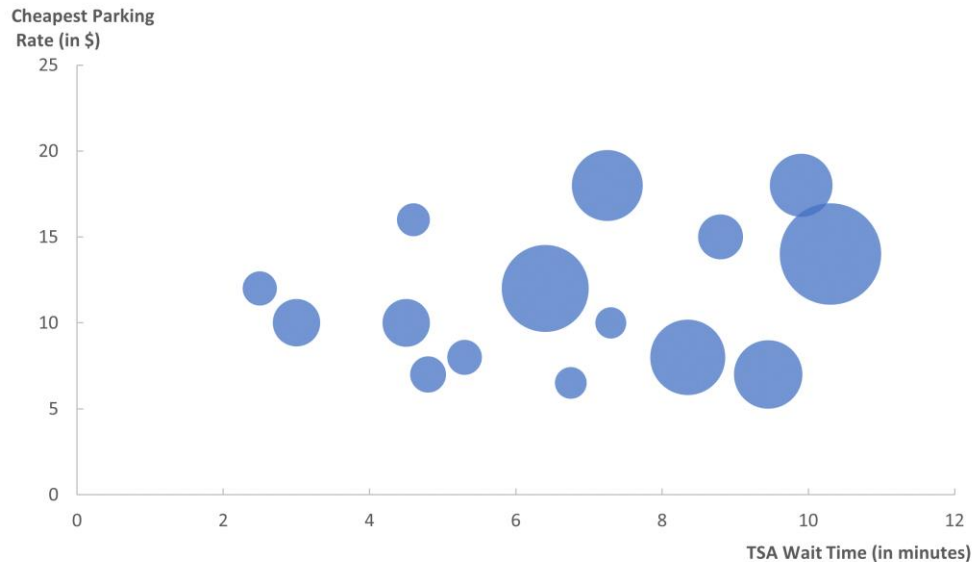
- Each point on the chart represents a pair of measurements for 51 U.S. cities.
- The measurements are average low temperature (in degrees Fahrenheit) and the average annual amount of snowfall (inches).
- The chart shows how the average annual amount of snowfall intuitively levels off at zero for warm-weather cities.
- Scatter charts are among the most useful charts for exploring pairs of quantitative data.



Building and Interpreting a Bubble Chart

	A	B	C	D
1	Airport Code	TSA wait time in minutes	Cheapest on-site parking cost in \$ per day	Annual Enplanements in millions of passengers
2	ATL	10.30	\$14.0	49.1
3	CLT	9.45	\$7.0	22.2
4	DEN	8.35	\$8.0	27.0
5	AUS	5.30	\$8.0	5.8
6	STL	4.80	\$7.0	6.2
7	SMF	7.30	\$10.0	4.6
8	RDU	6.75	\$6.5	4.8
9	EWR	9.90	\$18.0	18.8
10	SFO	7.25	\$18.0	24.0
11	LAX	6.40	\$12.0	36.1
12	SLC	4.50	\$10.0	10.8
13	SAN	8.80	\$15.0	9.7
14	MSY	4.60	\$16.0	5.3
15	IAD	3.0	\$10	10.7
16	BNA	2.50	\$12.0	5.6

Airport Passengers, TSA Wait Time and Cost Of Parking



- For each of the 15 airports, the following quantitative variables are shown:
 - Horizontal axis: Average wait time [in minutes] in the non-priority TSA queue.
 - Vertical axis: The cheapest on-site daily rate [in \$] for parking at the airport.
 - Bubble size: The number of enplanements [in millions of passengers] in a year, including transfers.

Editing a Bubble Chart

6	STL	
7	SMF	
8	RDU	
9	EWR	
10	SFO	
11	LAX	
12	SLC	
13	SAN	
14	MSY	
15	IAD	
16	BNA	
17		
18		
19		
20		
21		
22		

Airport Passengers, TSA Wait Time and Cost

Cheapest Parking
Rate (in \$)

25

20

15

10

5



Edit Series

Series name:

Select Range

Series X values:

= 10, 9, 8, 5, 5...

Series Y values:

= 14, 7, 8, 8, 7...

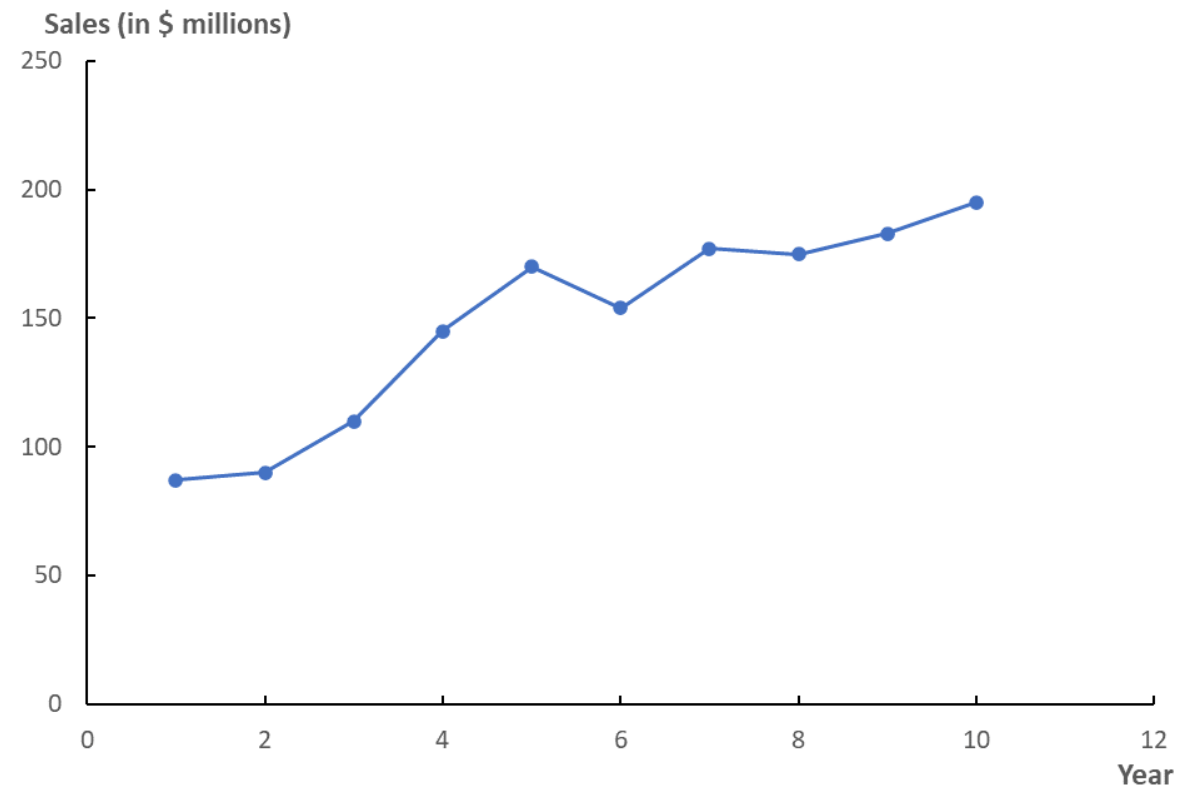
Series bubble size:

= 49.1, 22.2, ...

Line Charts

Line charts are good for time series data – the addition of lines connecting points suggests continuity and makes it easier to see and interpret changes over time.

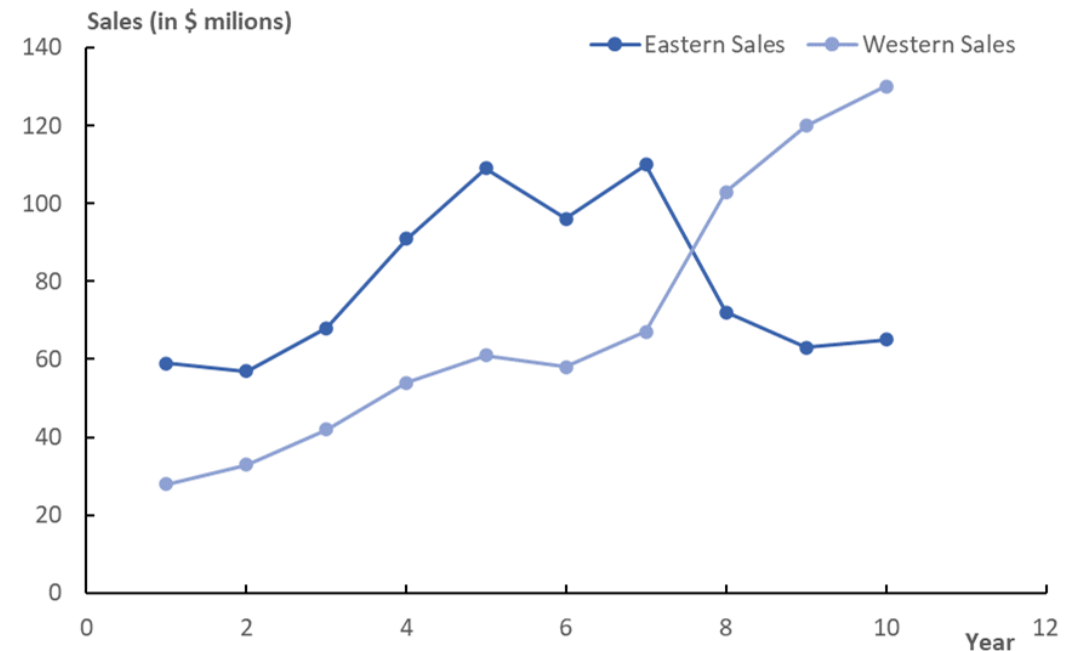
	A	B
1	Year	Sales (in \$ millions)
2	1	87
3	2	90
4	3	110
5	4	145
6	5	170
7	6	154
8	7	177
9	8	175
10	9	183
11	10	195



Multiple-Line Charts

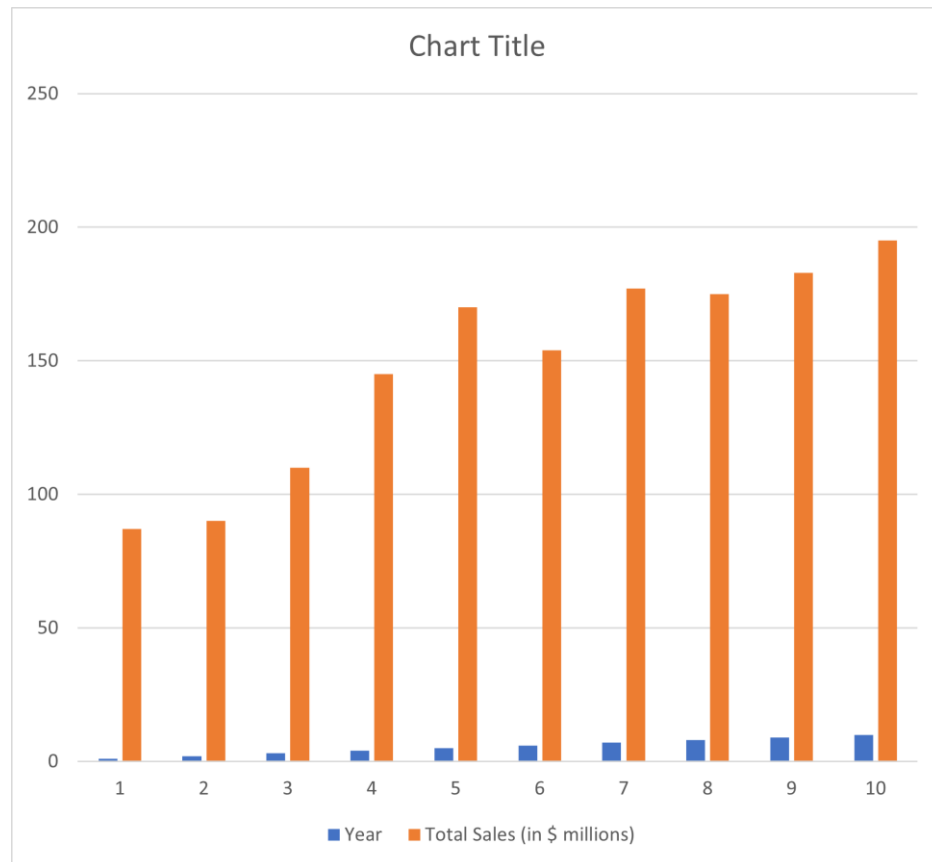
- Sales in the western region have increased over the last three of the given ten-year period.
- In contrast, sales in the eastern region have dropped significantly since year seven.

	A	B	C	D
1	Year	Eastern Sales	Western Sales	Total Sales (in \$ millions)
2	1	59	28	87
3	2	57	33	90
4	3	68	42	110
5	4	91	54	145
6	5	109	61	170
7	6	96	58	154
8	7	110	67	177
9	8	72	103	175
10	9	63	120	183
11	10	65	130	195

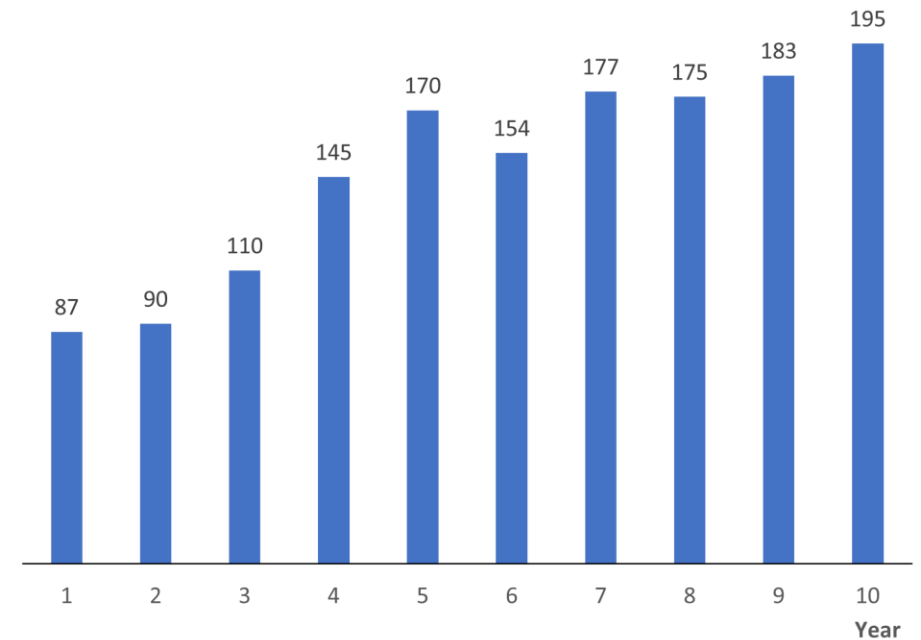


Column Charts

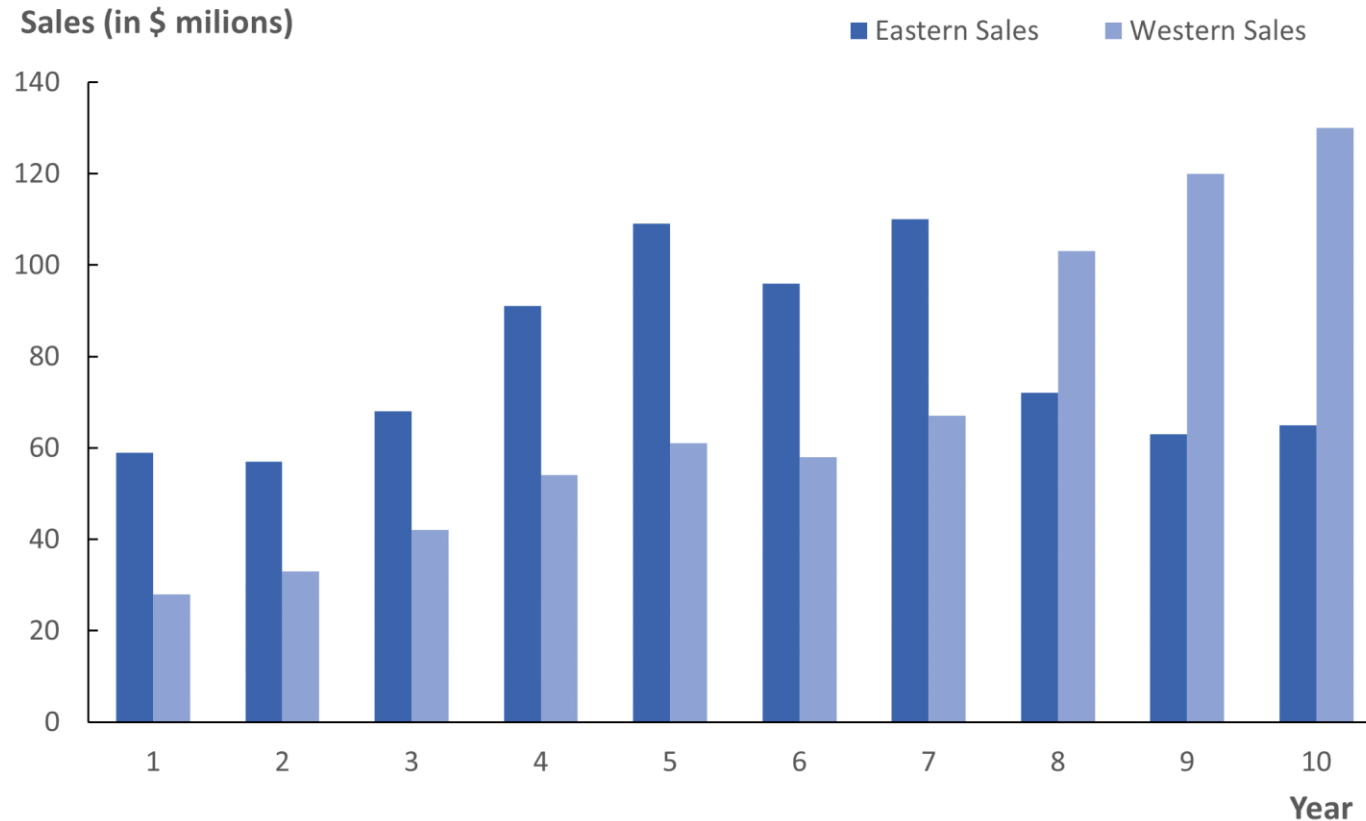
The chart on the left is incorrectly rendered because Excel displays the year as if it were a quantitative variable.



Total Sales (in \$ millions)

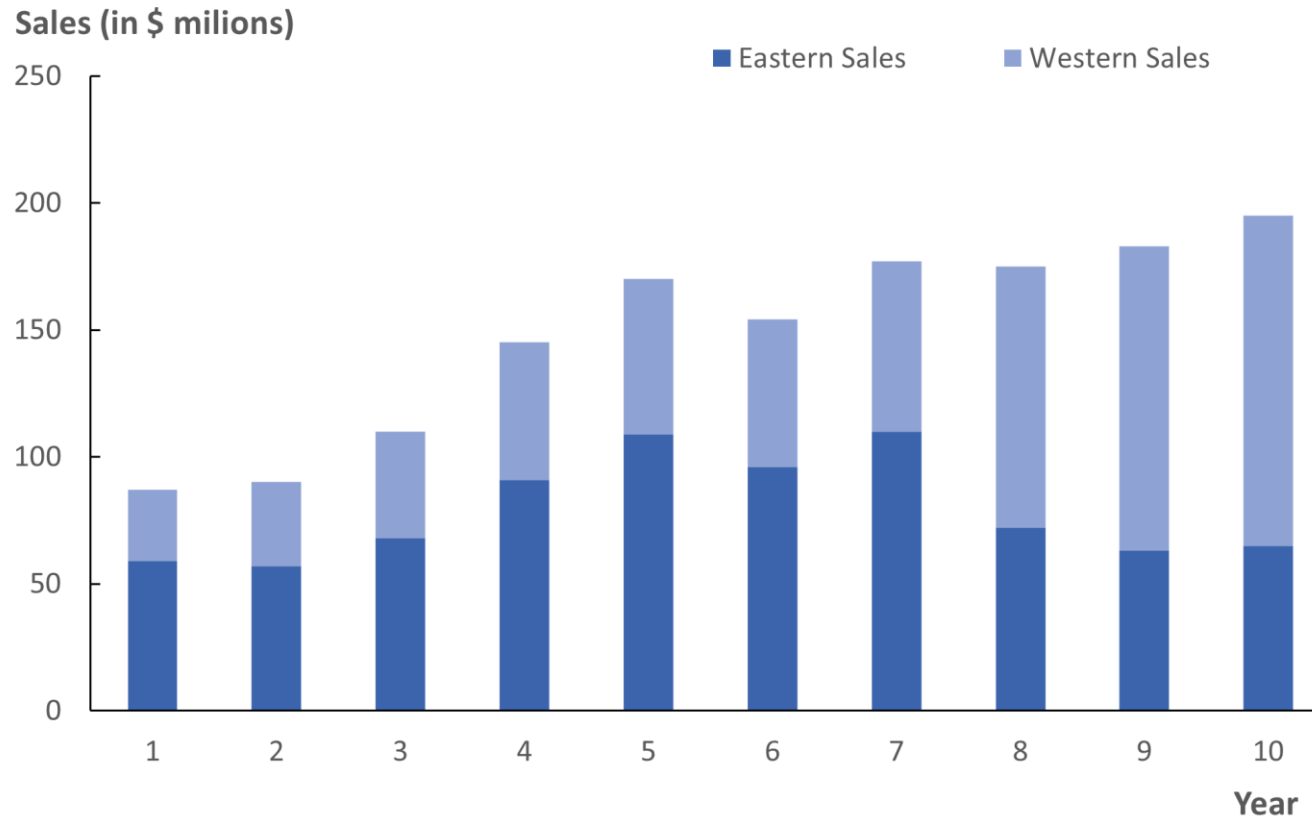


Clustered Column Charts



- The multiple line chart option is less cluttered than the clustered column chart.
- The best option to convey how total sales are changing is a stacked column chart.

Stacked Column Charts



This chart shows the combination of eastern and western sales by year and the total height of the column indicates total sales.

Bar Charts

- All column charts (clustered and stacked) can also be represented horizontally as bar charts.
- A bar chart may be preferable to a column chart when there are lengthy category names that are easier to display horizontally.
- For time series data, a column chart provides for a more natural display of the passage of time horizontally, from left to right.

End of Part 1